

GuardianEye

Deployment & User Guide

Version 1.0

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1. Architecture Overview

GuardianEye uses a client-server model over a single local area network. No internet connection or cloud services are required.

Component	Runs On	Purpose
GuardianEye.Admin	Teacher PC	Server, database, dashboard, reports
GuardianEye.Client	Student PC (each)	Lockdown agent, session timer, monitoring

[Diagram: Teacher PC ↔ Student PC 1, Student PC 2, ... via local network (UDP discovery + TCP communication)]

Communication Flow

- 1. **Discovery:** Admin broadcasts UDP beacons. Clients listen and connect to the admin IP.
- 2. **Login:** Student sends credentials via TCP. Admin validates against SQLite database.
- 3. **Session:** Timer counts down on client. Status updates sent every heartbeat.
- 4. **Commands:** Admin can force logout, add time, send chat, update filters in real time.

Network Requirements

Protocol	Port	Direction
UDP	50000	Broadcast (Admin → Clients, discovery)
TCP	50001	Bidirectional (all session traffic)

2. Deployment — Teacher PC (Admin)

System Requirements

- Windows 10 22H2 or Windows 11 (recommended for glass UI effects)
- .NET 10 Runtime (included in self-contained publish)
- No administrator privileges required for normal operation

Installation

1. Copy `GuardianEye.Admin.exe` to the teacher PC.
2. (Optional) Create a shortcut on the desktop.
3. Ensure Windows Firewall allows ports 50000 (UDP) and 50001 (TCP) — see Section 4.

First Launch

1. Run `GuardianEye.Admin.exe`.
2. The database is automatically created at `%LOCALAPPDATA%\GuardianEye\GuardianEye.db`.
3. Default groups and a sample student are seeded.
4. The admin server starts automatically (UDP beacon + TCP listener).

[Login Form — dark glass card on frosted backdrop, "GuardianEye Admin" title, two rounded glass inputs, glowing Login button]

3. Deployment — Student PCs (Client)

System Requirements

- Windows 10 22H2 or Windows 11
- .NET 10 Runtime (included in self-contained publish)
- Same local network as the teacher PC

Manual Installation

1. Copy `GuardianEye.Client.exe` to each student PC.
2. Run the executable. It will auto-discover the admin via UDP broadcast.

Auto-Start Installation (Recommended)

Run the following once to install the client as a startup application with a watchdog:

```
GuardianEye.Client.exe --install
```

This creates:

- A **Registry Run key** under `HKCU\Software\Microsoft\Windows\CurrentVersion\Run`
- A **Scheduled Task watchdog** that verifies the client is running every 5 minutes
- A **hidden copy** at `C:\Windows\Temp\GuardianEye\GE_Client.exe` for resilience

Removal

```
GuardianEye.Client.exe --remove
```

This deletes the Run key and the scheduled task. The hidden copy is deleted on next reboot by the watchdog cleanup.

The client self-protects against closure. Alt+F4, Task Manager, and process termination are blocked while the lock screen or session is active.

4. Firewall Configuration

GuardianEye uses two ports for communication. The Admin PC must allow inbound traffic on both.

Port	Protocol	Purpose	Open On
50000	UDP	Admin discovery beacon	Admin PC (inbound)
50001	TCP	Session traffic, chat, commands	Admin PC (inbound)

Windows Firewall (PowerShell — Run as Admin)

```
New-NetFirewallRule -DisplayName "GuardianEye UDP" -Direction Inbound -Protocol UDP -LocalPort 50000
New-NetFirewallRule -DisplayName "GuardianEye TCP" -Direction Inbound -Protocol TCP -LocalPort 50001
```

Third-Party Firewalls

If using a third-party firewall (McAfee, Norton, etc.), add exceptions for `GuardianEye.Admin.exe` for both UDP and TCP on port 50001.

Student PCs do not require any firewall changes — they initiate outbound TCP connections to the Admin.

5. First-Time Setup

Adding Students

Navigate to the Groups section from the dashboard toolbar.

1. Click **Groups** to open the group management panel.
2. Create a group (e.g., "Class A") with a daily time limit.
3. Students are managed individually — edit a student to assign them to a group.

[Group Management — dark glass panel with groups DataGridView, styled Add/Edit/Delete buttons, student list]

Configuring Filters

Click **Filters** to block websites and applications across all connected clients.

- Blocked Domains: one domain per line (e.g., `youtube.com`)
- Blocked Processes: one process name per line (e.g., `notepad`)

[Filter Configuration — dark glass panel with two multiline text areas, Apply and Close buttons]

Policy Export/Import

Click **Export** to save the complete policy (groups, students with settings, filters) as a JSON file. Click **Import** to restore or clone a policy to another Admin instance.

The import overwrites existing groups and student configurations. A confirmation prompt appears before proceeding.

6. Using the Admin Dashboard

[Dashboard — dark glass layout with stat cards (Total Students, Active Now, Today's Minutes), connected clients DataGridView with colored status indicators, activity log, action buttons, chat section at bottom]

Stat Cards

Three glass cards at the top display real-time statistics:

- **Total Students:** Number of connected clients
- **Active Now:** Clients with active sessions (green indicator)
- **Today's Minutes:** Total session minutes across all students today

Connected Clients Grid

Each row shows: Student Name, Username, Group, Remaining Time, Status (color-coded: Active=green, Idle=amber, Offline=red), Current Application, and action buttons.

- **Add Time:** Click to grant additional minutes to a specific student
- **Force Logout:** Immediately ends the student's session

Activity Log

A monospace log at the bottom records connections, disconnections, time requests, and admin actions with timestamps.

Chat

Send administrative messages to individual students or broadcast to all. Messages appear as toast notifications on student PCs (8-second auto-dismiss).

Reports

Click **Reports** to view session history, blocked attempts, and a summary tab with most-active students. Export sessions to CSV.

[Reports — dark TabControl with three tabs: Session History, Blocked Attempts, Summary. Each has a styled DataGridView.]

7. Student Session Experience

Lock Screen

When the client starts, it displays a full-screen lock overlay across all monitors. The lock screen blocks keyboard shortcuts, traps the mouse cursor, and cannot be closed by the student.

[Lock Screen — dark gradient background with large red text message, scanline overlay, glowing text effect]

Login

The student enters their username and password. After successful authentication, the lock screen is dismissed and the session timer form appears.

[Login Screen — centered glass card on frosted Mica backdrop, "GUARDIAN EYE" title, glass username/password inputs, blue Login button]

Session Timer

A sleek dark window displays the remaining time in large bold digits (60pt). A circular arc ring surrounds the timer, colored green (>50%), amber (25-50%), or red (<25%), with a subtle glow effect. Two buttons at the bottom allow the student to log out or request additional time.

[Session Timer — dark glass window with large "42:15" timer display, green circular arc ring, Logout and Request Time buttons]

Time Warnings

Popup warnings appear at **5 minutes remaining** and **2 minutes remaining** to alert the student.

Warnings

When time expires, or if the admin force-logouts, or the network disconnects, the session ends and the lock screen reappears. The student must log in again to continue.

Foreground App Monitoring

The admin dashboard displays the current foreground application for each connected student. For browsers (Chrome, Edge, Firefox, etc.), the page title is extracted and shown instead of the raw process name.

8. Feature Reference

Feature	Description
Student Login	Username/password authentication against local SQLite database
Session Timer	Countdown timer with configurable daily limit per student/group
Time Warnings	Popup alerts at 5 min and 2 min remaining
24/7 Lock Screen	Full-screen overlay across all monitors — blocks keyboard, traps cursor, immune to Alt+F4/Task Manager
Self-File-Lock	Client holds an open handle on its own EXE to prevent deletion while running
Watchdog	Scheduled task checks every 5 minutes that the client is running; restarts if missing
Hidden Copy	Resilience copy stored at C:\Windows\Temp\ with unique name
Browser Restriction	Whitelist: only Chrome browser is allowed; all other browsers blocked by process rule
Website Filtering	Real-time blocklist of domains (applied via host file entries)
Process Watchdog	Terminates blacklisted processes within 2 seconds
Network Watchdog	Locks screen and ends session if network connection is lost
Idle Monitor	Locks session after 3 minutes of inactivity
Foreground App Monitor	Reports current app to admin (with browser page title extraction)
Admin Chat	Send messages to individual students or broadcast to all
Time Requests	Student can request additional minutes with a reason; admin approves/denies
Force Logout	Admin can immediately end any student's session
Add Time	Admin can grant extra minutes to any student
Student Grouping	Organize students into groups with different daily limits

Reports & Analytics	Session history, blocked attempts, most-active students, CSV export
Remaining Time Persistence	Daily remaining time saved on logout and restored on next login (same day)
Policy Export/Import	Full policy (groups, students, filters) exportable as JSON for backup or cloning
Dashboard	Real-time client list with status indicators, stat cards, activity log, and chat
Glass UI	Windows 11 Mica backdrop, semi-transparent glass cards, rounded corners, glow animations, circular timer arc

Screenshot Guide

Run both applications on a test network to capture the following screenshots and insert them at the marked placeholder locations in this document:

#	Form	What to Capture
1	Admin Login	The dark glass login card centered on the frosted Mica backdrop
2	Admin Dashboard	Full dashboard with at least 2 students connected — show stat cards, grid, log, and chat
3	Group Management	Groups DataGridView with a few groups, student list visible
4	Filter Configuration	Filter form with sample domains and processes filled in
5	Reports	Session History tab with sample data
6	Client Login	Full-screen glass login card with "GUARDIAN EYE" title
7	Client Session	Timer window with the circular arc ring visible
8	Client Lock Screen	Full-screen overlay with the lock message and gradient background

Inserting Screenshots

Each placeholder in this document is marked:

```
div class="screenshot" [Description] /div
```

Replace the placeholder with an `<img>` tag:

```
<img src="screenshot_01.png" style="width:100%; border-radius:8px; box-shadow:0 4px 12px rgba(0,0,0
```